
AI Usage Declaration

Project: Super Mario Game (CS202 Final Project) **Group:** 52 **Members:** - Member A: Nguyễn Đình Minh Huy, 25125083 (Engine & Infrastructure) - Member B: Trần Gia Huy, 25125084 (Entities & Gameplay)

Statement of Usage

We declare that Artificial Intelligence (AI) tools were utilized during the development of this project. The AI acted as a pair-programming assistant, contributing to the following areas:

1. **Architecture & Design:** Suggesting and refining structural design patterns (e.g., State, Observer, Factory, Singleton, Memento, Object Pool).
2. **Implementation:** Assisting in writing boilerplate C++ code, engine subsystems, algorithms (e.g., AABB collision), and gameplay logic.
3. **Debugging & Refactoring:** Identifying bugs, fixing logic flaws, and optimizing code structure for maintainability.
4. **Documentation:** Generating report drafts, logging task history, and formatting class diagrams.

Tools Used

- **Antigravity** (Google, Gemini-based) — the primary agentic coding tool used across most of the project's history: implementing engine subsystems and design patterns, enemy/entity work, level content, weekly report drafting, and the majority of the git history's commits.
- **Claude Code** (Anthropic) — introduced later in the project, for the final push: diagnosing and fixing the three reported gameplay defects, implementing Endless Mode and the level-solvability oracle, writing and running the `verify_*` regression test suite additions, and generating/expanding this submission's documents (report, feature list, member contributions, testing checklist) directly from the codebase, git history and weekly reports rather than from a blank page.

Academic Integrity

While AI assisted in the development, we carefully reviewed, tested, and integrated all AI-generated code. We understand the underlying logic of the submitted system and take full responsibility for its correctness, architecture, and behavior. The custom game logic, level design, and system integration were actively directed, guided, and verified by the human developers.

Signatures:

_____ (Nguyễn Đình Minh Huy)

_____ (Trần Gia Huy)

Date: August 31, 2026